

Revised Generation Interconnection System Impact Study Report

For Queue Project Q16-0664

Airline 230 kV
July 31, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0664. The Interconnection Customer proposes a 381.4 MW natural gas combined cycle facility interconnecting at the Airline 230 kV substation in the Ohio area. The request was received on August 16, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 333 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 10 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$47,470,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Devil'S Slide-Lcent 230.0 kV L
Contingency: Airline-Omni-Hckberry 230.0 kV Line
Loading: 181.3% (Rating: 600 MVA)
DFAX: 66.2%
Pre-project loading: 139.2%
2. Facility: Devil'S Slide-Decorative Panels Intl Substation 230.0 kV
Contingency: Airline-Omni-Hackberry 230.0 kV line
Loading: 176.4% (Rating: 640 MVA)
DFAX: 66.2%
Pre-project loading: 136.9%
3. Facility: Airline-Decorative Panels Intl 230.0 kV
Contingency: Airline-Omni-Hackberry 230.0 kV
Loading: 170.0% (Rating: 702 MVA)
DFAX: 66.2%
Pre-project loading: 134.1%
4. Facility: Airline-Omni-Hackberry 230.0 kV
Contingency: Decorative Panels Intl-Airline 230.0 kV
Loading: 167.2% (Rating: 850 MVA)
DFAX: 82.6%
Pre-project loading: 130.1%
5. Facility: Sunnyvale-Dresden 230.0 kV
Contingency: Dresden-Conesville Sta 500.0 kV
Loading: 110.5% (Rating: 473 MVA)
DFAX: 6.4%
Pre-project loading: 105.4%

6. Facility: Omni-Hackberry-Lucent 230.0 kV
 Contingency: Airline-Decorative Panels Intl Substation 230.0 kV
 Loading: 109.5% (Rating: 653 MVA)
 DFAX: 19.3%
 Pre-project loading: 98.2%
7. Facility: W Canton-North Cambridge 230.0 kV
 Contingency: Wedpatch-Pt Arena 500.0 kV Line
 Loading: 107.5% (Rating: 902 MVA)
 DFAX: 5.7%
 Pre-project loading: 105.1%
8. Facility: Sunnyvale-Conesville Station 230.0 kV
 Contingency: Dresden-Conesville Sta 500.0 kV
 Loading: 107.4% (Rating: 459 MVA)
 DFAX: 6.4%
 Pre-project loading: 102.0%
9. Facility: Fishkill Plains-Dillard-Orange 345.0 kV line
 Contingency: Becker-Dismal River 345.0 kV
 Loading: 102.0% (Rating: 2,830 MVA)
 DFAX: 8.4%
 Pre-project loading: 100.9%
10. Facility: Wcanton 500-230 kV Autotransformer
 Contingency: Pt Arena-Weedpatch 500.0 kV
 Loading: 100.7% (Rating: 1,543 MVA)
 DFAX: 7.8%
 Pre-project loading: 98.8%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Devil'S Slide-Lcent 230.0 kV L	Rebuild line with higher-capacity conductor	\$8,752,000	89% (\$7,758,000)
Decorative Panels Intl-Devil'S Slide 230.0 kV	Rebuild line with higher-capacity conductor	\$6,338,000	89% (\$5,618,000)
Airline-Decorative Panels Intl 230.0 kV	Rebuild line with higher-capacity conductor	\$16,420,000	89% (\$14,554,000)
Omni-Hackberry-Airline 230.0 kV	Rebuild line with higher-capacity conductor	\$11,470,000	89% (\$10,184,000)
Sunnyvale Substation-Drsden 230.0 kV	Rebuild line with higher-capacity conductor	\$18,258,000	8% (\$1,370,000)
Omni-Hackberry-Lucent 230.0 kV Line	Reconductor line	\$4,378,000	74% (\$3,219,000)
W Canton-North Cambridge 230.0 kV	Reconductor line	\$5,131,000	10% (\$540,000)
Conesville Station-Sunnyvale 230.0 kV	Reconductor line	\$6,967,000	8% (\$523,000)
Fishkill Plains-Dillard-Orange 345.0 kV line	Reconductor line	\$51,960,000	4% (\$1,915,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Wcanton 500-230 kV Autotransformer	Replace transformer with higher-rated unit	\$28,632,000	6% (\$1,789,000)
	Total allocated to project		\$47,470,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.